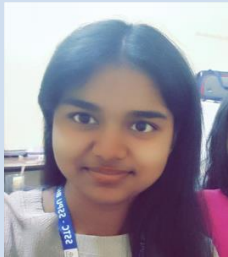


IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – WritePulse AI

Domain of Project – Media and Digital Content Creation

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Problem Statement -

Problem statement-Managing Content Overload for Blog Writers

The Challenge:

Bloggers today face an overwhelming flood of scattered information—news updates, research articles, social media trends, and audience feedback. This content is fragmented, quickly outdated, and difficult to organize, making it hard for writers to focus on creating impactful blogs. The challenge is not just to collect ideas but to synthesize, prioritize, and guide writers toward high-quality content creation.

The Objective:

Build an agentic AI application that acts as an intelligent writing assistant.

The solution should:

1. **Fuse Disparate Sources:** Collect real-time data from multiple channels (news sites, research papers, social media, comments) and synthesize into clear summaries with actionable blog topic suggestions.
2. **Enable Multimodal Content Curation:** Accept user inputs (text, images, or voice notes), refine ideas, and recommend blog structure, tone, or supporting visuals.
3. **Create a Predictive & Agentic Layer:** Analyze engagement patterns to predict trending topics and suggest optimal publishing time, hashtags, or angles for maximum reach.
4. **Visualize the Writing Pulse:** Provide a clean, real-time dashboard showing curated topics, sentiment analysis, and engagement forecasts to help writers align with audience interests.

Proposed Solution -

Proposed Solution: AI-Powered Content Curation & Blog Writing Agent

The proposed solution is a multi-agent AI-powered Content Assistant built using **IBM Bob** to provide real-time synthesis, trend forecasting, and guided content creation. The system addresses challenges in content overload, topic prioritization, and multimodal drafting through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate, real-time data from disparate channels such as news feeds, research repositories, and social trends. This data is embedded and stored in a vector database, enabling the system to retrieve, synthesize, and extract high-impact topics and insights using Granite models.

The system consists of four key agents:

Content Discovery & Synthesis Agent retrieves and summarizes multi-source information (news, academic papers, audience comments) using RAG to suggest high-impact blog topics.

Multimodal Curation & Structuring Agent processes user inputs (text briefs, reference images, voice memos) to refine ideas and generate tailored blog outlines, tones, and visual recommendations.

Trend & Reach Optimization Agent analyzes engagement patterns and sentiment to predict trending hooks, optimal publishing times, hashtags, and distribution angles.

Draft & Feedback Agent assists writers in drafting sections, evaluating readability, checking consistency, and delivering actionable editorial feedback.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes curated topic clusters, sentiment analytics, and audience engagement forecasts to guide editorial workflow.

IBM Bob is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and tone adaptation.

Overall, this solution delivers an **end-to-end intelligent writing assistant** that streamlines the research-to-drafting lifecycle, eliminating content overload and maximizing reader engagement.

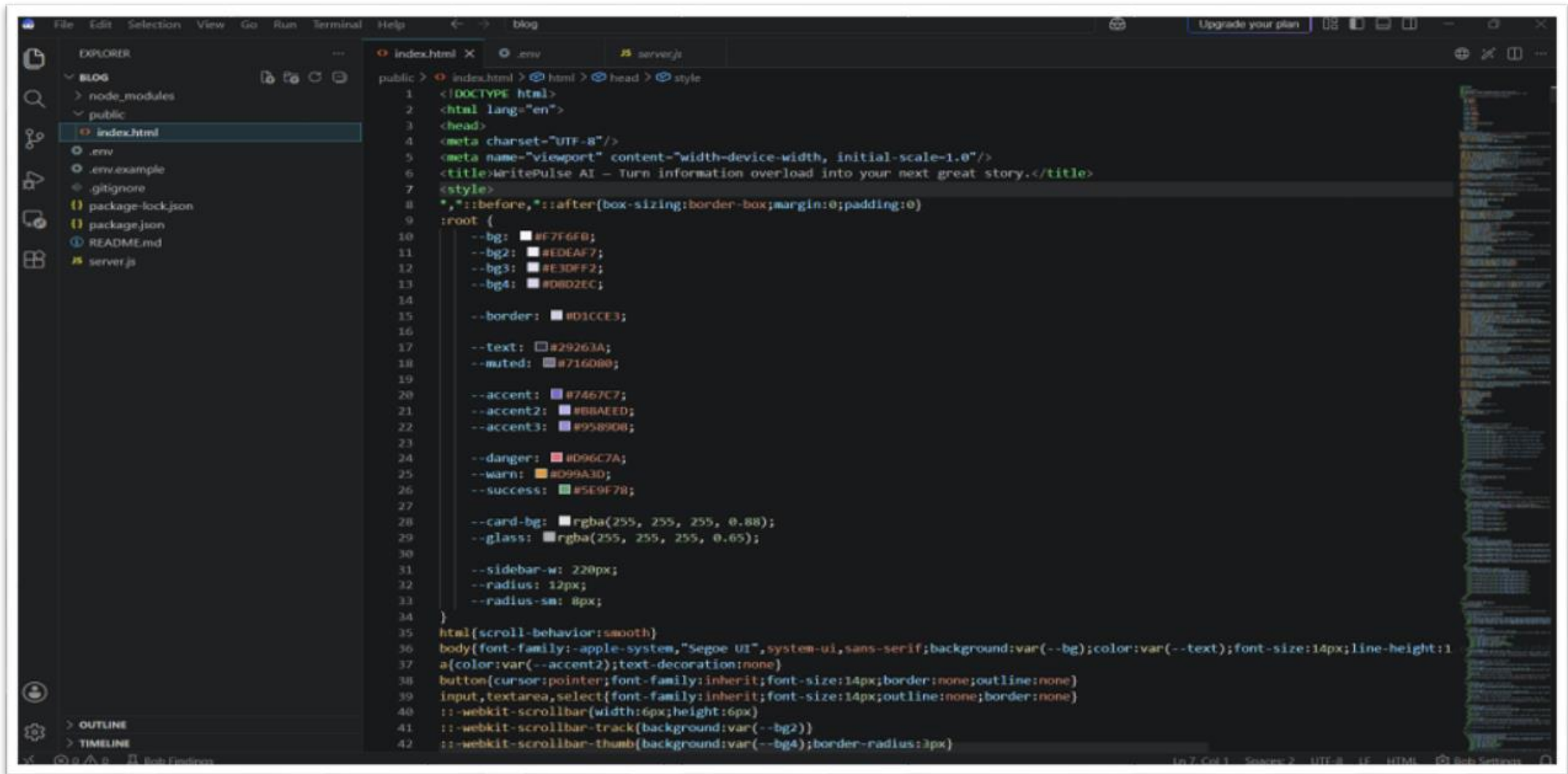
Technology Used

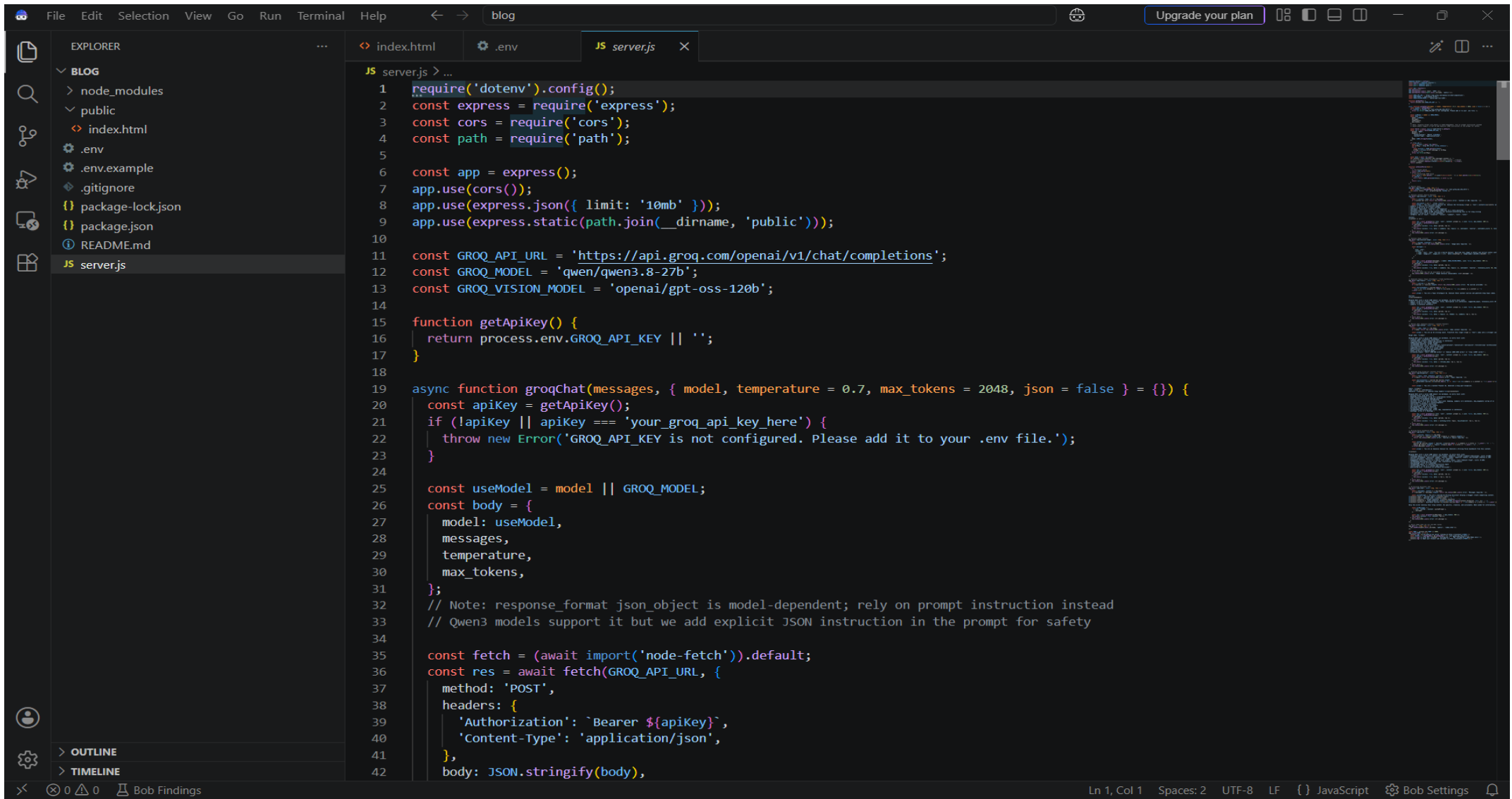
1. **IBM Bob** – AI-assisted application development and agentic workflow implementation
2. **Groq API + Llama Model** – Natural language understanding, reasoning, summarization, topic generation, and personalization
3. **HTML5 + CSS3 + JavaScript** – Single-page responsive frontend and interactive dashboard
4. **Node.js + Express** – Secure backend/API layer for Groq integration
5. **RAG** – Retrieval of relevant real-time/source content before AI generation
6. **News/Research APIs** – Real-time content and trend collection
7. **LocalStorage** – Saving drafts, topics, sources, and user preferences
8. **Data Visualization** – Sentiment, trends, and engagement forecast visualization

Required Files created using IBM bob -

- 1) index.html
- 2) .env
- 3) Server.js
- 4) package.json
- 5) package-lock.json

IBM Bob Screenshots:





Role of Agentic AI in the solution

Role of Agentic AI in Blog Writing Agent

Agentic AI enables the Content Assistant to function as an intelligent, autonomous system rather than a simple text generator. Using IBM watsonx.ai and IBM Granite models, it coordinates multiple specialized agents to deliver end-to-end writing support. Each agent performs a specific role—such as retrieving real-time web trends, structuring multimodal user notes, predicting audience engagement, and assisting with draft generation—while working collaboratively. This multi-agent approach ensures efficient task execution and accurate results.

Agentic AI also brings context awareness, understanding the writer's preferred niche, target audience persona, and tone to generate tailored outlines and content hooks. It supports real-time adaptation, continuously refining suggestions based on live feedback, breaking news, and multimodal inputs like voice notes and reference images. Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like an autonomous editorial partner that helps creators overcome content overload and craft high-impact blogs.



Project Output Screenshot

Project Output Screenshot

WritePulse AI

Turn information overload into your next great story.

Overview

Content Inbox

Topic Studio

Idea Refinery

Blog Blueprint

Writing Pulse

AI Writer

Groq API connected

WritePulse AI v1.0

Content Inbox

Collect information from any source. Everything becomes a smart content card.

Add Content

Text

URL

Image

Voice

TITLE / LABEL

E.g. Tech article, comment, idea...

CATEGORY

Personal Note

CONTENT

Paste news, research, comments, ideas...

Add & Analyze

Content Cards (0)

Search...

All Types

Clear All

No content yet

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 Blog Blueprint

Generate a complete blog structure from any topic.

Configure Blueprint

TOPIC / TITLE

Enter topic or select from Topic Studio

WRITING TONE

 Informative

TARGET AUDIENCE

E.g. Tech enthusiasts, beginners...

☒ Use collected sources as context

No blueprint yet

Enter a topic and click **Generate Blueprint**.

Or select a topic from the Topic Studio.

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WritePulse AI v1.0

🔥 AI Writing Assistant

Develop your blog content with an intelligent writing partner.

Current Context

TOPIC

Current topic...

TONE

Informative

AUDIENCE

Who are you writing for?

Quick Actions

Stronger introduction

Alternative hooks

More conversational

Add counterargument

Better title

What am I missing?

Write conclusion

Clear Chat

WritePulse Writing Assistant

No topic selected

Ready

Novelty and Uniqueness

The proposed WritePulse AI stands out by combining Agentic AI, content synthesis, multimodal input, trend analysis, and personalized writing assistance into a single intelligent platform for blog writers.

Agentic Content Intelligence – Instead of functioning as a basic chatbot, specialized AI modules work together to analyze sources, identify trends, prioritize topics, and guide the writer from idea discovery to blog creation.

Multi-Source Content Fusion – Brings together information from news, research articles, social trends, audience comments, and user-provided content, transforming scattered information into meaningful summaries and actionable insights.

AI-Powered Topic Discovery – Identifies emerging themes and generates relevant blog topics, unique angles, titles, and content opportunities from the collected information.

Multimodal Idea Refinement – Allows writers to provide ideas through text, images, or voice notes and transforms rough inputs into refined content concepts, recommended tones, structures, and supporting visuals.

Predictive Content Intelligence – Analyzes available trend and engagement signals to estimate topic momentum, audience interest, suitable publishing times, hashtags, and potential engagement.

Writing Pulse Dashboard – Provides a real-time visual overview of trending topics, sentiment, topic momentum, engagement forecasts, and recommended content directions, allowing writers to make data-informed decisions.

Personalized Writing Workflow – Adapts recommendations according to the writer's selected topic, target audience, preferred tone, and collected sources, creating a more relevant and efficient content-generation experience.

From Information Overload to Content Creation – Unlike conventional writing assistants that mainly generate text, WritePulse AI focuses on the complete journey: collect → analyze → synthesize → prioritize → plan → write.

This makes WritePulse AI an intelligent, proactive, and data-driven content companion, helping writers transform scattered information into meaningful and engaging blog content rather than simply generating text on demand.

Git Hub Link

[reddylmanasha-cmd/blog: AI powered content intelligence platform for smarter blog ideation , planning and writing](#)

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Future Scope

1. Advanced Social Media & Trend Integration –

The system can be integrated with real-time social media platforms and trend sources to monitor emerging topics, hashtags, audience discussions, and viral content. This will help WritePulse AI identify trending subjects earlier and provide more relevant blog ideas, angles, and content recommendations.

2. AI-Powered Voice Assistant & Multilingual Support –

Future versions can include a voice-enabled assistant and support for multiple languages, allowing writers to capture ideas through voice notes and interact with the system in their preferred language. This will make the platform more accessible and useful for bloggers from diverse linguistic backgrounds.

3. Personalized Audience & Engagement Prediction –

The system can be enhanced to learn from a writer's previous content and audience interactions to predict which topics, writing styles, and publishing times may generate higher engagement. This can enable more personalized and data-driven content planning.

4. Automated Content & Visual Curation –

Future versions can automatically recommend relevant images, infographics, charts, references, and other supporting visuals based on the selected blog topic and structure, reducing the time writers spend searching for supporting content.

Certificates Earned:



Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

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(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 10 Sep 2026 (GMT)

Learning hours: 50 mins

Certificates Earned:

9/9/26, 2:39 PM

Completion Certificate | SkillsBuild

IBM SkillsBuild

Completion Certificate



This certificate is presented to

L Manasha Reddy

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 09 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.